

Trainee Program

Deshmukh Technologies Trainee Program

A 90-day full-stack software development and professional development program that prepares trainees to build, test, debug, explain, document, and maintain production software.

PRIMARY OUTCOME

Junior Full-Stack Developer

BACKEND TRACKS

Java / Spring Boot or Python / FastAPI

DELIVERY

Git + GitHub + Docker + CI/CD + AWS

FRONTEND

React + TypeScript

DATABASE

MySQL / PostgreSQL

DURATION

90 Days · 12 Phases

Learn. Build. Solve. Review. Deploy.

Grow.

INTERNAL TALENT DEVELOPMENT
STANDARD

DOCUMENT DTTP Program Handbook	VERSION 1.1	CLASSIFICATION Internal Training Standard	OWNER Deshmukh Technologies
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PART I Program Foundation

01 Program Purpose

The Deshmukh Technologies Trainee Program (DTTP) is a structured professional development program designed to transform freshers, interns, graduates, and entry-level candidates into capable software engineers.

The program combines technical education, practical coding, full-stack development, project-based learning, engineering practices, mentorship, code review, testing, debugging, security, DevOps, professional development, and continuous assessment.

The program does not define success as completion of 90 days. Success means the trainee can demonstrate the ability to build, test, debug, explain, document, and maintain software.

02 Program Vision

Build skilled, responsible and production-ready software professionals who can understand business requirements, develop complete applications, solve problems independently and contribute effectively to real Deshmukh Technologies projects.

03 Program Mission

1. Establish strong programming fundamentals.
2. Develop modern full-stack development skills.
3. Teach professional software engineering practices.
4. Provide practical project experience.
5. Develop independent problem-solving ability.
6. Build communication and teamwork skills.
7. Introduce testing, security and deployment.
8. Measure capability continuously.
9. Provide a structured path from trainee to developer.
10. Create a repeatable internal talent-development system.

04 Three-Dimensional Development Model

Every trainee is evaluated across three dimensions. A trainee must develop all three.

A. Technical

- Programming
- Frontend
- Backend
- Database
- APIs
- Testing
- Security
- Cloud

B. Engineering

- Git
- Architecture
- Clean code
- Debugging
- Code review
- CI/CD
- Docker
- Documentation

C. Professional

- Communication
- Discipline
- Teamwork
- Ownership
- Time management
- Accountability
- Problem solving

05 Technology Architecture

COMMON FOUNDATION

PROGRAMMING

OOP

DATA STRUCTURES

ALGORITHMS

WEB

SQL

GIT / GITHUB

REST APIS

FRONTEND

HTML

CSS

JAVASCRIPT

TYPESCRIPT

REACT

ENGINEERING

TESTING

SECURITY

DOCKER

CI/CD

AWS

MONITORING

BACKEND — TRACK A

JAVA

SPRING BOOT

SPRING DATA JPA

HIBERNATE

BACKEND — TRACK B

PYTHON

FASTAPI

SQLALCHEMY

06 Trainee Journey

01 Application	02 Screening	03 Technical Assessment
04 Interview	05 Selection	06 Orientation
07 Baseline Assessment	08 Foundation	09 Frontend
10 Backend Specialization	11 Database & APIs	12 Full-Stack Project
13 Testing	14 Security	15 Docker
16 CI/CD	17 AWS	18 Final Project
19 Final Assessment	20 Production Readiness	21 Junior Full-Stack Developer

07 90-Day Master Roadmap

Phase	Days	Outcome
1	1–5	Professional and engineering foundation
2	6–15	Programming and problem solving
3	16–25	Web and frontend foundation
4	26–35	React + TypeScript
5	36–45	Java / Spring Boot or Python / FastAPI
6	46–55	Database + REST API
7	56–65	Full-stack project
8	66–72	Authentication + security
9	73–78	Testing + debugging
10	79–82	Docker
11	83–86	CI/CD + AWS
12	87–90	Final project + assessment

08 Training Methodology

Recommended allocation of trainee time:

Theory		20%
Practical		30%
Project		40%
Review & Assessment		10%

The trainee should spend more time building software than listening to someone explain software.

09 Daily Development Cycle



This should become the trainee’s normal development habit.

PART II Technical Curriculum

10 Phase I — Orientation

Days 1–5. Orientation establishes company context and professional standards before technical training begins.

COMPANY

- Deshmukh Technologies
- Vision and mission
- Products and services
- Projects and teams
- Development methodology
- Customer expectations

PROFESSIONAL STANDARDS

- Attendance and working hours
- Communication and meetings
- Workplace conduct
- Confidentiality
- Information security
- Responsibility

11 Baseline Assessment

Before technical training begins, assess the trainee and record a **Baseline Skill Profile**.

PROGRAMMING

- Logic
- Variables
- Conditions
- Loops
- Functions

TECHNICAL

- Java
- Python
- JavaScript
- SQL
- Git

PROBLEM SOLVING

- Logical reasoning
- Coding problem

COMMUNICATION

- Self-introduction
- Technical explanation
- Problem explanation

12 Git & GitHub

Git is mandatory. Trainees must learn repository, clone, commit, branch, push, pull, merge, rebase, stash, revert, pull request, code review, and conflict resolution.

STANDARD WORKFLOW

TASK	BRANCH	CODE	TEST	COMMIT	PUSH	PULL REQUEST	REVIEW	FIX	APPROVE	MERGE
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13 Phase 2 — Programming

Days 6–15.

Variables	Data types	Operators	Conditions	Loops	Functions	Collections	Strings	Exceptions	OOP
Data structures	Algorithms								

BEGINNER EXERCISES

- Reverse string
- Palindrome
- Fibonacci
- Factorial
- Prime
- Largest element

INTERMEDIATE EXERCISES

- Two Sum
- Anagram
- Missing number
- Duplicate detection
- Frequency counting
- Sorting, searching, LinkedList

14 Java Track

JAVA FUNDAMENTALS

- OOP, classes, interfaces
- Inheritance, polymorphism, encapsulation
- Exceptions
- Collections
- Generics

MODERN JAVA

- Lambda
- Streams
- Optional
- Functional interfaces
- Date/time API

15 Spring Boot

CONTROLLER	SERVICE	REPOSITORY	JPA / HIBERNATE	DATABASE
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Spring Boot	REST	Dependency Injection	DTO	Entity	Validation	Exception handling	Logging	JPA
Hibernate	Spring Security	JWT	Testing					

16 Python Track

Variables Lists Tuples Sets Dictionaries Functions Modules Packages Exceptions File handling

OOP Type hints Generators Decorators Async

17 FastAPI

ROUTER SERVICE REPOSITORY SQLALCHEMY DATABASE

Routes Request models Response models Pydantic Validation Dependency injection Middleware

Exceptions Authentication Async API documentation

18 Frontend — React + TypeScript

Trainees learn components, props, state, hooks, forms, validation, routing, protected routes, API integration, error handling, loading states, and responsive design.

```
src/  
├─ components/  
├─ pages/  
├─ layouts/  
├─ services/  
├─ hooks/  
├─ types/  
├─ utils/  
├─ routes/  
└─ assets/
```

19 Database

Trainees learn SQL, tables, keys, constraints, relationships, joins, indexes, transactions, normalization, and query optimization.

ORM: Java uses JPA / Hibernate. Python uses SQLAlchemy.

ORM knowledge must not replace SQL knowledge.

20 REST API

Both backend tracks implement the same business API so Java and Python trainees learn the same API design principles.

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate user
GET	/api/employees	List employees
POST	/api/employees	Create employee
GET	/api/employees/{id}	Get employee
PUT	/api/employees/{id}	Update employee
DELETE	/api/employees/{id}	Delete employee
GET	/api/departments	List departments
POST	/api/departments	Create department
GET	/api/attendance	List attendance
POST	/api/attendance	Record attendance
GET	/api/leaves	List leaves
POST	/api/leaves	Apply for leave
PUT	/api/leaves/{id}/approve	Approve leave

PART III Project & Engineering

21 Major Training Project

Employee Management System

Roles: ADMIN · HR · EMPLOYEE

Authentication	Dashboard	Employee Management	Department Management	Attendance	Leave	Reports
Profile						

22 Full-Stack Architecture

JAVA IMPLEMENTATION

REACT	REST	SPRING BOOT	SERVICE	JPA / HIBERNATE	MYSQL / POSTGRESQL
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PYTHON IMPLEMENTATION

REACT	REST	FASTAPI	SERVICE	SQLALCHEMY	MYSQL / POSTGRESQL
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Both implementations must satisfy the same business requirements.

23 Authentication & Security

Authentication

Authorization

Password hashing

JWT

Roles

Permissions

CORS

Input validation

SQL injection awareness

XSS fundamentals

Secrets management

Environment variables

HTTPS

Least privilege

Never commit passwords, API keys, tokens, private keys, or production credentials to GitHub.

24 Testing

Backend

Java: JUnit, Mockito, Spring Boot tests

Python: Pytest, FastAPI testing, mocking

Frontend

- Component testing
- Form testing
- API-state testing
- Error-state testing

API

- Postman
- Positive and negative tests
- Authentication and authorization
- Edge cases

25 Debugging

BUG

REPRODUCE

READ ERROR

LOGS

NETWORK

API

BACKEND

DATABASE

ROOT CAUSE

FIX

RETEST

REGRESSION

The trainee should learn to determine which layer is responsible rather than randomly changing code.

26 Docker

FRONTEND CONTAINER

BACKEND CONTAINER

DATABASE CONTAINER

Learn Dockerfile, image, container, network, volume, environment variables, and Docker Compose.

27 CI/CD

DEVELOPER

GITHUB

BUILD

LINT

UNIT TESTS

INTEGRATION TESTS

DOCKER BUILD

ARTIFACT

DEPLOYMENT

HEALTH CHECK

28 AWS

IAM

EC2

S3

RDS

CloudFront

Route 53

Security Groups

CloudWatch

The trainee should understand the purpose of each service before using it.

29 Project Management

Every trainee project should use a professional work breakdown.

Level	Example
Epic	Employee Management
Features	Authentication, Employees, Departments, Attendance, Leave, Reports
User Story	As an administrator, I want to create an employee so that the employee can be managed in the system.
Task	Create employee POST API.
Subtasks	Entity, DTO, Repository, Service, Controller, Validation, Test

This teaches trainees how professional teams convert requirements into engineering work.

PART IV Assessment & Career

30 Definition of Done

A feature is complete only when all of the following are true:

☐ Requirement understood

☐ Acceptance criteria satisfied

☐ UI implemented

☐ API implemented

☐ Database completed

☐ Validation added

☐ Security checked

☐ Tests written

☐ Tests passed

☐ Error cases checked

☐ Git commit created

☐ Pull Request created

☐ Code review completed

☐ Review comments resolved

☐ Documentation updated

☐ Mentor approved

31 Mentor System

Each trainee should have **1 Primary Mentor** and, where practical, **1 Technical Reviewer**.

MENTOR KPIS

Trainee progress

Code-review quality

Feedback quality

Skill-gap identification

Project completion

Production-readiness improvement

This prevents mentorship from becoming informal or inconsistent.

32 Daily Report

Field	Question
Daily Objective	What was planned?
Completed	What was actually completed?
Learning	What was learned?
Problems	What went wrong?
Investigation	What was checked?
Resolution	How was it fixed?
Git Activity	Commits / PRs
Next Plan	Tomorrow's objectives

33 Weekly Review

The mentor reviews technical progress, assignment completion, project progress, code quality, problem solving, communication, discipline, Git activity, blockers, and skill gaps.

34 Competency Model

Level	Definition
L0	Not exposed
L1	Understands
L2	Performs with guidance
L3	Performs independently — graduation target for core skills
L4	Can review / explain
L5	Can own

Advanced areas such as architecture, CI/CD, and AWS may initially target **L2**.

35 Performance Status

 Green On track.	 Amber Needs improvement.	 Red At risk.
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Amber and Red trainees receive an **Individual Improvement Plan**.

36 Weekly Score

Area	Weight
Technical Learning	15
Practical Work	20
Problem Solving	15
Project Contribution	20
Code Quality	10
Testing	5
Communication	5
Discipline	5
Documentation	5
Total	100

37 Final Assessment

Category	Weight
Programming	15%
Frontend	10%
Backend	15%
Database / API	10%
Final Project	20%
Problem Solving	10%
Testing / Debugging	5%
Git / CI/CD	5%
Documentation	5%
Communication / Professionalism	5%
Total	100%

38 Graduation Levels

85–100 • Developer Ready

Eligible for Junior Full-Stack Developer responsibilities.

70–84 • Junior Developer — Supervised

Can contribute with closer mentoring.

55–69 • Extended Training

Requires additional structured training.

Below 55 • Management Review

Formal decision required.

39 Production Readiness Gate

Before graduation, the trainee must demonstrate:

☐ Requirement understanding

☐ System design

☐ React development

☐ Backend development

☐ SQL

☐ REST APIs

☐ Authentication

☐ Authorization

☐ Testing

☐ Debugging

☐ Git

☐ Pull Requests

☐ Code review

☐ Docker

☐ CI/CD understanding

☐ Deployment understanding

☐ Documentation

☐ Technical communication

40 Career Progression

TRAINEE

FOUNDATION TRAINEE

TECHNICAL TRAINEE

PROJECT TRAINEE

JUNIOR FULL-STACK DEVELOPER

SOFTWARE DEVELOPER

SENIOR DEVELOPER

TECHNICAL LEAD

Promotion must be based on **demonstrated capability**, not simply completion of the 90-day period.

41 DTTP Management System

The long-term goal is an internal Trainee Management System.

ORGANIZATION DASHBOARD

- Total trainees
- Active trainees
- Completed
- Extended
- At-risk

INDIVIDUAL RECORD

- Attendance and daily reports
- Assignments and technical scores
- Git activity, PRs, and code reviews
- Project progress and skill matrix
- Mentor rating and final result

PROFILE

BASELINE

ATTENDANCE

DAILY REPORTS

ASSIGNMENTS

WEEKLY REVIEWS

SKILL MATRIX

PROJECT

CODE REVIEWS

FINAL ASSESSMENT

READINESS

GRADUATION

42 Final DTTP Standard

The program should not ask: “How many technologies did the trainee complete?”

What can this trainee independently build, test, debug, explain, document and maintain?

STUDENT

TRAINEE

PRACTITIONER

PROJECT CONTRIBUTOR

INDEPENDENT TRAINEE

PRODUCTION-READY

JUNIOR FULL-STACK DEVELOPER

DESHMUKH TECHNOLOGIES

DTP — Deshmukh Technologies Trainee Program

Learn. Build. Solve. Review. Deploy. Grow.

A disciplined, technically capable and production-
ready Junior Full-Stack Developer.